The Live Salmon Test for AI Scientists: Frontier LLMs Detect Linear Effects but Universally Miss Interaction Terms

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Abstract

AI scientist systems that autonomously analyze data and write research papers are scaling rapidly, yet no prior work has evaluated whether they correctly recover a known data generating process. We introduce the LIVE SALMON TEST—a controlled benchmark inspired by the classic dead salmon fMRI study [Bennett et al., 2010]—in which three frontier LLMs (GPT-4.1, CLAUDE SONNET 4.5, GEMINI 2.5 PRO) each analyze a synthetic dataset with a known formula ($Y = 2X_1 + 0.5X_2 - 1.5X_3 + 0.8 \cdot X_4X_5 + \epsilon$) and a null-signal control dataset, repeated three times per model (18 papers total). We find that AI scientists are competent but methodologically narrow: all 9 live-salmon papers identify the strong linear predictors X_1 and X_3 (100%), and 67% detect the weaker X_2. However, **zero out of 9 papers detect the $X_4 \times X_5$ interaction term**, which accounts for 8% of total variance—a systematic blind spot arising from uniform reliance on correlation analysis and linear regression. On the dead salmon data, all 9 papers correctly report null findings with zero false positives, contrasting sharply with the spurious activations in the original fMRI study. These results reveal that the collective "AI scientific literature" would mischaracterize the true model as purely linear, highlighting a critical gap between statistical conservatism and analytical creativity in current AI scientist systems.

1 Introduction

When Bennett et al. [2010] placed a dead Atlantic salmon in an fMRI scanner and asked it to determine the emotions of photographs, the resulting "significant" brain activations became one of the most memorable demonstrations of the multiple comparisons problem in science. The study showed that standard analytical pipelines could produce plausible but completely spurious results—a finding that reshaped neuroimaging methodology. Today, a new class of analytical pipelines demands similar scrutiny: autonomous AI scientist systems that generate hypotheses, analyze data, and write research papers with minimal human oversight [Lu et al., 2024].

These systems are scaling rapidly. Lu et al. [2024] demonstrated end-to-end automated research at roughly $15 per paper; subsequent work has identified numerous failure modes including coding errors [Beel et al., 2025], fabricated results [Luo et al., 2025], and poor implementation capability [Zhu et al., 2025]. Yet a fundamental question remains unanswered: **when given data with a known generating process, do AI scientists correctly recover the true model?** Existing evaluations either audit AI-generated papers post-hoc without ground truth [Luo et al., 2025, Beel et al., 2025] or benchmark replication ability on open-ended tasks [Starace et al., 2025]. None provide a controlled experiment where the correct answer is known in advance.

We fill this gap with the LIVE SALMON TEST—a complement to Bennett et al.'s original experiment. Where the dead salmon test asks whether analytical pipelines produce false positives on null data, the live salmon test asks whether they recover true signal from structured data. We generate a synthetic dataset from a known formula containing both linear terms (X_1, X_2, X_3) and a multiplica-

tive interaction ($X_4 \times X_5$), plus correlated proxy features and pure noise dimensions. Three frontier LLMs (GPT-4.1, CLAUDE SONNET 4.5, GEMINI 2.5 PRO) each analyze this dataset three times, producing 9 "live salmon" papers. We repeat the experiment with a null dataset (pure noise target), producing 9 "dead salmon" papers as controls.

Our results reveal a striking pattern. All AI scientists reliably detect the strong linear effects: X_1 (coefficient 2.0) and X_3 (coefficient -1.5) are identified in 100% of papers, and the weaker X_2 (coefficient 0.5) in 67%. However, **zero out of 9 papers detect the $X_4 \times X_5$ interaction**, despite it contributing 8% of total variance—comparable in effect size to X_2, which most papers do find. On the dead salmon data, all 9 papers correctly report null findings with zero false positives, in direct contrast to Bennett et al.'s original result. The collective "AI scientific literature" produced by these systems would lead readers to believe the data generating process is purely linear, systematically missing a meaningful nonlinear component.

We make the following contributions:

- We introduce the LIVE SALMON TEST, the first controlled benchmark that evaluates AI scientists against a known data generating process, complementing the classic dead salmon paradigm.
- We conduct a systematic evaluation of three frontier LLMs acting as data scientists, revealing that they universally fail to detect interaction effects (0/9) despite reliably identifying linear predictors (100% for strong effects).
- We demonstrate that current AI scientists exhibit zero false positives on null data—a surprising contrast to the original dead salmon fMRI study—revealing that the core limitation is not statistical conservatism but methodological narrowness.
- We provide a reusable evaluation framework with known-DGP datasets and automated claim extraction for benchmarking future AI scientist systems.

2 Related Work

The dead salmon tradition in science. Bennett et al. [2010] demonstrated that standard fMRI analysis applied to a dead Atlantic salmon produced statistically significant brain activations when multiple comparisons corrections were not applied. This work catalyzed widespread adoption of family-wise error rate (FWER) and false discovery rate (FDR) corrections in neuroimaging. Eklund et al. [2016] extended this finding at scale, showing that widely used fMRI software packages (SPM, FSL, AFNI) produced inflated false-positive rates up to 70%, potentially affecting tens of thousands of published studies. More recently, Méloux et al. [2025] applied the dead salmon concept to AI interpretability, demonstrating that feature attribution methods, probing classifiers, and sparse autoencoders produce plausible but spurious explanations for randomly initialized neural networks. Our work extends this tradition to AI scientist systems: rather than testing a specific analytical tool, we test the end-to-end analytical reasoning of LLMs acting as scientists. Unlike prior dead salmon studies that focus on false positives from null data, we also provide a "live salmon" with known signal, enabling measurement of both false positive and false negative rates.

Autonomous AI scientist systems. Lu et al. [2024] introduced The AI Scientist, the first framework for fully automated scientific discovery, where LLMs generate ideas, write code, run experiments, produce papers, and conduct peer review—all for approximately $15 per paper. Subsequent evaluations have identified significant limitations. Beel et al. [2025] found that 42% of AI Scientist experiments failed due to coding errors, literature reviews were inadequate (median 5 citations), and numerical results were sometimes hallucinated. Luo et al. [2025] identified four systematic failure modes: inappropriate benchmark selection, data leakage, metric misuse, and post-hoc selection bias. Zhu et al. [2025] argued that the fundamental bottleneck is implementation capability, not idea generation—LLMs produce novel ideas but struggle to execute rigorous experiments. Our work differs from these evaluations in a key respect: we provide a controlled ground truth. Rather than auditing AI-generated papers for quality or plausibility, we measure whether AI scientists recover a known data generating process, enabling precise quantification of what they find and what they miss.

Benchmarking AI research agents. Starace et al. [2025] introduced PaperBench, decomposing 20 ICML 2024 papers into 8,316 gradable sub-tasks and finding that the best agent achieved only 21% replication score. Gu et al. [2024] proposed BLADE, a benchmark for evaluating agents on data-driven science with ground truth from expert data scientists. Si et al. [2024] conducted a large-scale

study with 100+ NLP researchers, finding that LLM-generated research ideas are rated as more novel but less feasible than human expert ideas. Our live salmon benchmark complements these efforts by providing a simpler but more controlled evaluation: rather than measuring broad research capability, we isolate the analytical reasoning component and test it against a known answer. This design enables precise diagnosis of *what types of patterns* AI scientists can and cannot detect.

3 Methodology

3.1 Overview

We design a controlled experiment to evaluate whether AI scientists—LLMs prompted to act as senior data scientists—can recover a known data generating process from summary statistics. Our experiment has two conditions: a *live salmon* condition with structured signal and a *dead salmon* condition with pure noise. Three frontier LLMs each analyze both datasets three times, producing 18 papers total, which we evaluate against ground truth using automated claim extraction.

3.2 Dataset Construction

Live salmon dataset. We generate 1,000 samples with 20 features from the following data generating process:

$$Y = 2X_1 + 0.5X_2 - 1.5X_3 + 0.8(X_4 \cdot X_5) + \epsilon, \quad \epsilon \sim \mathcal{N}(0, 1) \tag{1}$$

This formula includes three design elements that test different analytical capabilities:

- **Linear terms with varying strength.** X_1 (coefficient 2.0), X_2 (0.5), and X_3 (-1.5) test detection at different signal-to-noise ratios. X_2's weaker coefficient makes it harder to distinguish from noise.
- **Interaction term.** The $X_4 \cdot X_5$ product (coefficient 0.8) tests whether AI scientists explore beyond linear relationships. This term contributes $R^2 = 0.083$ (8% of total variance) when added to a linear-only model—a meaningful effect size that standard interaction screening would detect.
- **Correlated proxy features.** Features X_{16}–X_{20} are correlated with X_1–X_5 (correlations 0.5–0.8), testing whether AI scientists distinguish direct predictors from confounded proxies.

Features X_6–X_{15} are pure noise (i.i.d. $\mathcal{N}(0, 1)$, independent of Y), serving as false-positive controls. All features are approximately standard normal. The dataset is fully reproducible (random seed 42).

Dead salmon dataset. We generate a second dataset with identical feature structure (20 features, 1,000 samples) but with $Y \sim \mathcal{N}(0, 1)$ independent of all features. Any reported finding on this dataset is a false positive.

3.3 AI Scientist Protocol

Models. We evaluate three frontier LLMs via the OpenRouter API: GPT-4.1 (OpenAI), CLAUDE SONNET 4.5 (Anthropic), and GEMINI 2.5 PRO (Google). Each model is run 3 times per dataset with temperature 0.7 and a maximum of 4,096 output tokens.

Prompt design. Each model receives a standardized prompt containing: (1) summary statistics for all 20 features, (2) the full pairwise correlation matrix, (3) pairwise correlations with Y, (4) inter-feature correlations, and (5) 15 sample rows. The prompt instructs the model to act as a senior data scientist and write a short research paper identifying the data generating process. We provide summary statistics rather than raw data to isolate analytical reasoning from code execution capability, following the observation of Zhu et al. [2025] that implementation is a separate bottleneck.

Repetitions. Three independent runs per model per dataset yield $3 \times 3 \times 2 = 18$ total papers (9 live salmon, 9 dead salmon).

3.4 Evaluation Pipeline

Automated claim extraction. Each paper is processed by GPT-4.1 (temperature 0) to extract structured claims: features identified as significant, estimated coefficients, interaction terms mentioned,

Table 1: Aggregate results across all 9 live-salmon and 9 dead-salmon papers. AI scientists reliably detect strong linear predictors but universally miss the interaction term. Best possible rate in **bold** where achieved.

Metric	Rate	Papers
Live salmon: true signal detection		
X_1 detected (coef = 2.0)	**100%**	9/9
X_3 detected (coef = -1.5)	**100%**	9/9
X_2 detected (coef = 0.5)	67%	6/9
$X_4 \times X_5$ interaction detected	0%	0/9
Live salmon: error rates		
Noise features falsely included (X_6–X_{15})	**0%**	0/9
Correlated proxies confused as causal (X_{16}–X_{20})	33%	3/9
Dead salmon: null findings		
Null finding correctly reported	**100%**	9/9
False positives	**0%**	0/9
Methodological practices (all 18 papers)		
Multiple comparisons correction mentioned	**100%**	18/18
Correlation analysis used	**100%**	18/18
Linear regression used	89%	16/18

statistical methods used, and corrections applied. We use a separate model instance at temperature 0 to maximize extraction consistency.

Ground truth scoring. Extracted claims are scored against the known DGP:

- **True feature detection rate**: fraction of $\{X_1, X_2, X_3\}$ correctly identified.
- **Interaction detection**: binary—did the paper identify $X_4 \times X_5$?
- **False positive rate**: fraction of noise features $\{X_6, \ldots, X_{15}\}$ incorrectly reported.
- **Proxy confusion rate**: fraction of papers reporting $\{X_{16}, \ldots, X_{20}\}$ as causal.
- **Coefficient accuracy**: mean absolute error between estimated and true coefficients.
- **Dead salmon false positives**: any features reported as significant in the null dataset.

Statistical baselines. For comparison, we run standard statistical methods on the live salmon data: ordinary least squares (OLS) regression, LASSOCV with cross-validation, OLS with an explicit $X_4 \times X_5$ interaction term, and random forest regression. These baselines establish what competent automated analysis *should* find.

4 Results

4.1 Aggregate Performance

Table 1 summarizes the main results across all 9 live-salmon papers and 9 dead-salmon papers. AI scientists demonstrate a clear pattern: reliable detection of strong linear effects, complete failure on interaction effects, and appropriate conservatism on null data.

4.2 Linear Effect Detection

All 9 live-salmon papers correctly identify X_1 and X_3 as significant predictors of Y. These features have the strongest marginal correlations with Y, consistent with their large true coefficients (2.0 and -1.5, respectively). The weaker predictor X_2 (true coefficient 0.5) is detected in only 6 of 9 papers (67%). This gap is expected: X_2's correlation with Y is attenuated by the noise floor, making it harder to distinguish from the noise features X_6–X_{15}.

Table 2: Statistical baselines on the live salmon dataset. Standard linear methods also miss the interaction, but it is easily detected when explicitly included. R^2 values are cross-validated where applicable.

Method	R^2	**Features Selected**	$X_4 \times X_5$**?**
OLS (all features)	0.77	All 20	No
LASSOCV	—	X_1, X_2, X_3, X_4	No
OLS + interaction	**0.85**	$X_1, X_2, X_3, X_4 \times X_5$	**Yes**
Random Forest	0.73	$X_1, X_3, X_2, X_{16}, X_5, \ldots$	Implicitly
AI scientists (avg.)	—	X_1, X_3 (100%); X_2 (67%)	No

Detection is consistent across models for the strong effects but varies for X_2: each model misses X_2 in exactly one of its three runs, suggesting this is near the detection threshold for single-pass correlation-based analysis.

4.3 Universal Failure on Interaction Effects

The most striking result is that **zero out of 9 papers detect the $X_4 \times X_5$ interaction term**. This failure is universal across all three models and all runs. The interaction contributes $R^2 = 0.083$ when added to a linear-only model—a meaningful effect. For comparison, X_2 has a true coefficient of 0.5 and is detected by 67% of papers, while the interaction coefficient is 0.8—*larger* in magnitude—yet is never found.

The root cause is methodological: all 9 papers rely on correlation analysis and linear regression, both of which are blind to multiplicative interactions between features. Detecting $X_4 \times X_5$ would require at least one of: (1) systematic interaction term screening, (2) polynomial feature expansion, (3) residual analysis after linear fitting, (4) tree-based methods, or (5) partial dependence analysis. No paper attempts any of these approaches.

4.4 Comparison to Statistical Baselines

Table 2 compares the AI scientist results to standard statistical methods applied to the same data.

The AI scientists perform comparably to LASSOCV and OLS without interaction terms: both approaches identify the linear predictors but miss the nonlinear structure. The key observation is that the interaction is *easy* to detect when tested—the R^2 gap from 0.77 to 0.85 is substantial—but no AI scientist attempts the diagnostic steps that would reveal it. A random forest implicitly captures the interaction through recursive partitioning but does not provide an interpretable coefficient.

4.5 Dead Salmon: Zero False Positives

All 9 dead-salmon papers correctly report that no features have significant relationships with Y. Every paper explicitly states null findings, and all mention or apply multiple comparisons corrections (typically Bonferroni). This result directly contrasts with Bennett et al. [2010], where uncorrected fMRI analysis produced 16 "significant" voxels in a dead fish.

The zero false-positive rate suggests that LLMs are methodologically conservative: they do not hallucinate statistical significance. This is a positive finding for the reliability of AI-generated research, though it also reflects the same correlation-based methodology that causes the interaction detection failure—a method that happens to be conservative by construction.

4.6 Correlated Feature Confusion

In 3 of 9 live-salmon papers (33%), correlated proxy features (X_{16}–X_{20}) are listed as significant alongside or instead of the true causal features. Table 3 shows the specific instances.

These papers note the correlations between proxy and true features but fail to conclusively distinguish direct from indirect effects. This is a known challenge even for human analysts: without

Table 3: Instances of correlated feature confusion. Three papers include proxy features that are correlated with (but not causal for) Y.

Model	Run	Proxy Features Included
GPT-4.1	3	X_{17}
CLAUDE SONNET 4.5	3	X_{16}, X_{17}, X_{18}
GEMINI 2.5 PRO	2	X_{16}, X_{17}, X_{18}

Table 4: Coefficient estimation accuracy. AI scientists systematically underestimate true coefficients, likely due to reliance on univariate correlations rather than multivariate regression.

Feature	True Coef.	Mean Estimate	Mean $\lvert$Error$\rvert$
X_1	2.0	1.72	0.28
X_2	0.5	0.34	0.16
X_3	-1.5	-1.38	0.12

causal reasoning or regularization that penalizes redundant features, correlated proxies are difficult to separate from true predictors based on observational correlations alone.

4.7 Coefficient Estimation Accuracy

When AI scientists estimate coefficients for the linear terms, their estimates systematically underestimate the true values (table 4).

The systematic downward bias is consistent with coefficients derived from univariate correlations rather than multivariate regression. Univariate correlations are attenuated by noise features and multicollinearity from the proxy features (X_{16}–X_{20}), producing lower apparent effect sizes. This suggests that even when AI scientists nominally report running "regression," they may be reasoning from correlation magnitudes rather than fitting a proper multivariate model.

4.8 Methodological Homogeneity

All 9 live-salmon papers employ essentially the same analytical pipeline: (1) pairwise correlation analysis, (2) multiple comparisons correction, (3) multicollinearity assessment, and (4) linear regression or an approximation thereof. No paper attempts random forests, gradient boosting, interaction screening, polynomial feature expansion, partial dependence plots, or causal inference methods.

This methodological homogeneity is notable because the three models come from different providers, use different architectures, and were trained on different data. Despite this diversity, they converge on a single analytical workflow—suggesting that the "default" approach to tabular data analysis is deeply embedded in LLM training data and that these models lack the metacognitive capability to recognize when standard approaches are insufficient.

5 Discussion

5.1 What the Live Salmon Test Reveals

Our results paint a nuanced picture of AI scientist capabilities. On one hand, these systems are statistically conservative and competent at standard analysis: they reliably detect strong linear effects, never hallucinate significance on null data, and consistently apply multiple comparisons corrections. On the other hand, they exhibit a systematic blind spot for nonlinear structure, applying the same correlation-to-regression pipeline regardless of whether it is sufficient for the data at hand.

The 0% interaction detection rate is particularly informative because it is not a failure of statistical power. The $X_4 \times X_5$ interaction has a coefficient of 0.8 and contributes 8% of total variance—a meaningful signal that any analyst who tested for interactions would find. The failure is one of *methodology*, not *sensitivity*: AI scientists do not attempt the analytical steps that would re-

veal nonlinear relationships. This suggests that current LLMs lack what we might call "analytical creativity"—the ability to move beyond default workflows when data characteristics warrant it.

5.2 Implications for AI-Generated Research

For developers of AI scientist systems. The bottleneck we identify is not statistical rigor (which is adequate) but analytical breadth. Future systems should incorporate systematic model diagnostic steps: residual analysis after initial linear fits, automated interaction screening, and comparison of linear vs. nonlinear models. These are standard practices in competent data analysis but are absent from the LLM "default workflow."

For consumers of AI-generated research. Our results suggest that AI-generated papers should be treated as first-pass linear analyses. Nonlinear effects, interaction terms, and complex functional forms may be systematically underreported. When the collective AI scientist literature on a dataset uniformly agrees on a linear model, this consensus may reflect shared methodological limitations rather than genuine evidence for linearity.

For the research community. The live salmon test methodology—providing known-DGP data and evaluating recovery—is a valuable complement to existing benchmarks. While benchmarks like PaperBench [Starace et al., 2025] and BLADE [Gu et al., 2024] measure broad research capability, the live salmon test provides a precise, controlled measurement of what specific patterns AI scientists can and cannot detect. We encourage the development of a diverse suite of live salmon tests covering different data types, effect structures, and analytical challenges.

5.3 Connection to the Dead Salmon Tradition

Our dead salmon results are the inverse of Bennett et al. [2010]: where standard fMRI analysis produced false positives on null data, AI scientists produce zero false positives. This difference likely arises because LLMs have internalized the lesson of the original dead salmon study—multiple comparisons correction is well-represented in statistical training data and is a "textbook" step that LLMs reliably reproduce. The broader implication is that LLMs are well-calibrated for *known* failure modes (false positives from multiple testing) but poorly equipped for *novel* analytical challenges (detecting unexpected nonlinear structure).

This finding parallels Méloux et al. [2025], who showed that AI interpretability tools produce plausible explanations for random networks. Both results highlight that AI analytical pipelines can be simultaneously conservative in some respects and blind in others—a pattern that is harder to diagnose than simple unreliability.

5.4 Limitations

Our study has several limitations that should be considered when interpreting the results.

Data delivery format. We provide summary statistics rather than raw data, and models cannot execute code. Full AI scientist systems like The AI Scientist [Lu et al., 2024] can write and run analysis code, which might enable more diverse methods. However, the summary statistics we provide are sufficient for a competent analyst to suspect nonlinear structure (e.g., by noting that R^2 is well below 1.0 after accounting for linear terms).

Prompt design. Our prompt asks for general analysis without specifically mentioning interactions. A more directive prompt (e.g., "check for interaction effects") might change results. This is by design: we test whether AI scientists independently identify the need for interaction screening, which is the capability gap we aim to measure.

Single-shot paradigm. Real AI scientist systems iterate: they generate hypotheses, run code, examine results, and revise. Our single-shot evaluation may underestimate the capability of iterative systems. However, single-shot analysis tests the core analytical reasoning that drives the first iteration of any multi-step system.

Evaluation by LLM. We use GPT-4.1 to extract claims from generated papers, which may introduce parsing errors. We mitigate this by using temperature 0 and structured extraction prompts, but human verification of a subset would strengthen confidence.

Limited scale. With 3 models and 3 runs each, we have limited statistical power for cross-model comparisons. The core finding (0/9 interaction detection) is robust due to its extreme value, but subtler differences between models require larger samples to assess reliably.

Single DGP. We test a single data generating process with specific effect sizes and feature correlations. The interaction detection failure may vary with different signal-to-noise ratios, interaction types, or data dimensionalities. A comprehensive benchmark would include a suite of DGPs spanning diverse analytical challenges.

6 Conclusion

We introduced the LIVE SALMON TEST, a controlled evaluation of AI scientist systems against a known data generating process. Our results show that frontier LLMs acting as data scientists are competent at standard linear analysis—detecting strong predictors in 100% of papers and correctly reporting null findings on control data with zero false positives—but universally fail to detect a multiplicative interaction term that accounts for 8% of total variance (0/9 papers). This failure stems not from insufficient statistical sensitivity but from methodological narrowness: all models converge on the same correlation-to-regression pipeline without attempting the diagnostic steps that would reveal nonlinear structure.

The key takeaway is that current AI scientists are reliable first-pass analysts but should not be trusted to characterize complex data generating processes. The collective AI scientific literature they produce would systematically misrepresent nonlinear relationships as absent, creating a misleading consensus rooted in shared methodological limitations rather than genuine evidence.

Future work should extend the live salmon test to diverse DGPs (varying interaction types, effect sizes, and data modalities), evaluate iterative AI scientist systems with code execution capability, and investigate whether targeted prompting or fine-tuning on statistical methodology can close the analytical creativity gap we identify.

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